

HFImage F4GOP

User manual

Digital images over HF — HFImage modes and EasyPal / QSSTV (HamDRM)

Version 0.82

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github.com/f4gopdenis-cyber/HFImage_F4GOP

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1. Overview

HFImage F4GOP sends and receives **digital pictures over HF radio**. One program covers two families of modes, and it **always listens to both at the same time**:

- **HFImage modes (DATAC)**: a modern OFDM modem (codec2) with **fountain-code** error correction. The received picture is **bit-for-bit identical** to the one sent, or it is not shown at all: never a damaged picture. Both stations must run HFImage.
- **EasyPal mode (HamDRM)**: the digital SSTV mode used by **EasyPal** and **QSSTV**. HFImage exchanges pictures, texts and hybrid pictures with stations running those programs, so this mode can live on now that EasyPal is no longer updated since the death of its author, Erik Sundstrup VK4AES.

Main features

- Sharp 800×600 photo in about 2 minutes (fast DATAC1X mode, 2.2 kHz wide).
- **Auto** mode: mode and robustness adapt to the reports of the stations.
- EasyPal / QSSTV compatible mode: modes A / B / E, QAM4 / 16 / 64, 2.2 or 2.4 kHz, Reed-Solomon, BSR / FIX, hybrid mode.
- Transmission **follows the received mode**: answer each station in the mode it uses.
- **QRM cutter** on receive: carriers, CW, RTTY and adjacent SSB signals removed.
- **Stations online** list, **Rx Ok** with reports in dB, **map** of stations.
- **Hybrid** mode (picture via the Internet, link over the air), including **flash hybrid**: a 4.5 s announcement over the air.
- Up to **4 correspondents' pictures** inserted into yours, animated GIFs, **QSL** card, text in the waterfall.
- **Separate windows**: waterfall, gallery, station list… placed where you want, even on another screen.
- More than 300 radios controlled (Hamlib), Kenwood direct, OmniRig, FLRig.
- **Gallery**, upload to your website (**Digi-Sites**), reception **analyser**.
- Automatic updates, 5 languages (English, French, German, Spanish, Italian).



Main window: status panel and buttons on the left, waterfall, stations and Rx Ok lists, received picture, history and received messages.

2. Installation and updates

Installation

1. Download **HFImage_F4GOP.exe** from the *Releases* page of the GitHub project.
2. Put it in a folder of your own, for example **C:\Radio\HFImage** (avoid “Program Files”).
3. On first start, Windows SmartScreen may show a warning: click “More info”, then “Run anyway”.

Nothing else to install: a single file, Windows 8.1 / 10 / 11 64-bit. For **Windows 7 SP1 or 8** (64-bit), take **HFImage_F4GOP_Win7.exe** from the same page: same program, updated with its own version; only SVG pictures cannot be displayed there.

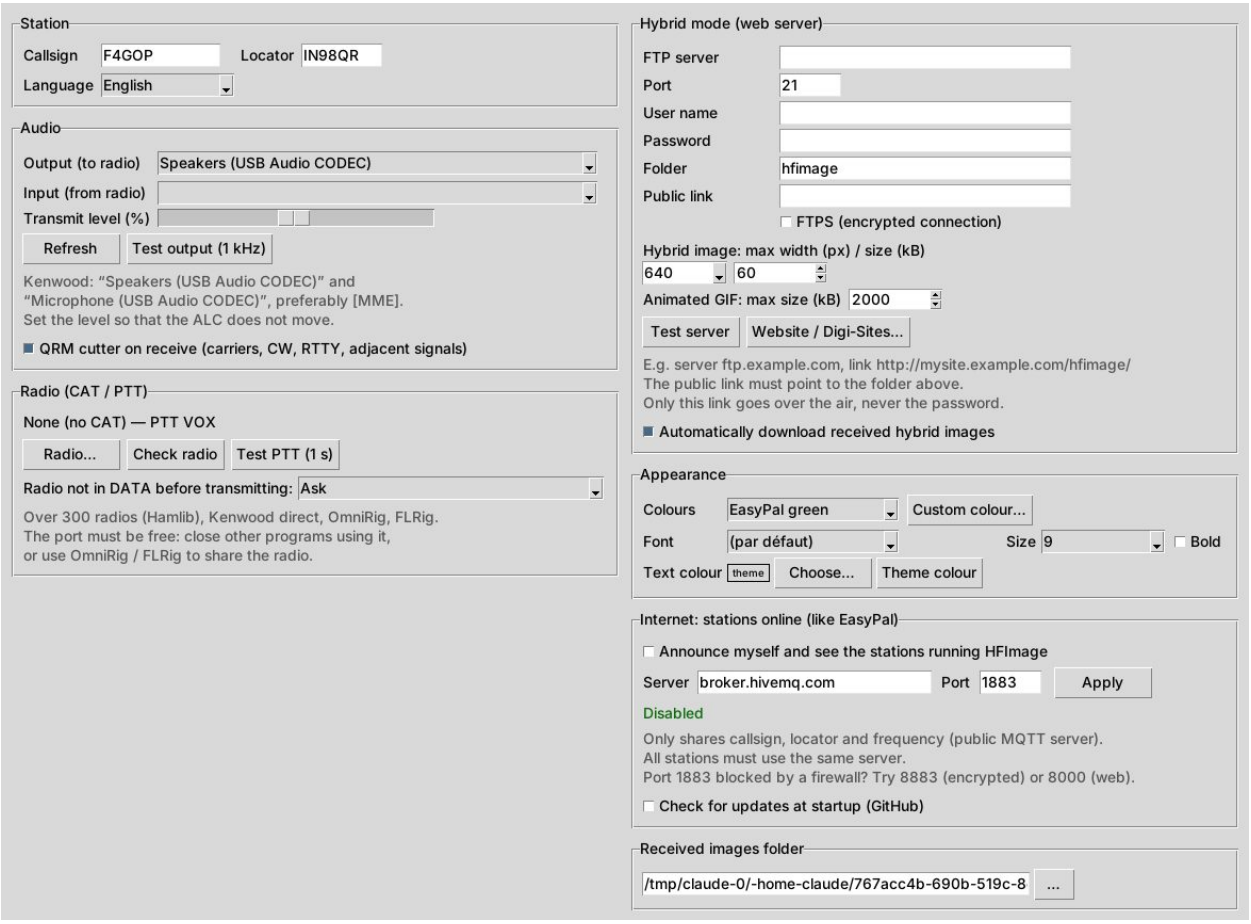
Automatic updates

At start-up (then every 3 hours), HFImage checks GitHub for a newer version. If there is one, an **orange banner** appears at the top of the window: one click, “Yes”, and the program updates itself and restarts. Menu **Info → Check for updates** to check at any time. The list of changes of every version is on the GitHub page (CHANGELOG).

> **Tip:** after an update, if Windows Explorer still shows the old icon, restart the PC (Windows caches icons).

3. First setup

Open **Settings** (button or menu). Settings are saved automatically.



The Settings window.

Station

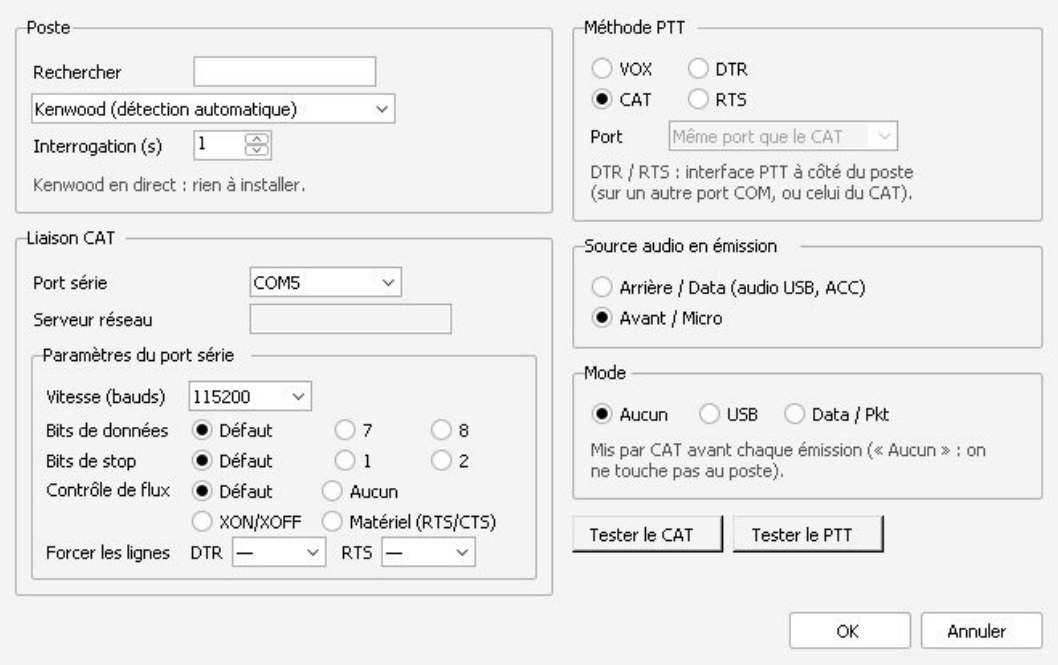
Callsign and **locator** (6 characters, e.g. IN98QR): used in the announcements, the station map and the QSL card. Interface language: English, Français, Deutsch, Español, Italiano.

Audio

- **Output (to radio)** and **Input (from radio)**: with a Kenwood TS-890 / TS-990, “Speakers (USB Audio CODEC)” and “Microphone (USB Audio CODEC)”. Each sound card appears only once, with its full name: HFImage picks the most reliable Windows interface itself and tries the others automatically if it fails.
- **Test output (1 kHz)**: 2 s tone without PTT, to check the sound card.
- **Transmit level**: set it so that the **ALC does not move**. An over-driven OFDM signal becomes unreadable.
- **QRM cutter on receive**: keep it ticked (see chapter 10).

Radio (CAT / PTT)

Radio… button: the window follows the Radio tab of WSJT-X.



Radio settings: rig, CAT link, PTT, audio source, mode, tests (screenshot of the French interface).

- **Radio**: type part of the name in “Search” (e.g. 7300, FT-991, TS-990). Kenwood TS-990S / TS-890S / TS-590 / TS-2000 / TS-480 are driven directly, nothing to install. Other radios: Hamlib, downloaded once (“Install Hamlib” button, 3 MB).
- **OmniRig** (Rig 1 / Rig 2) or **FLRig**: to share the radio with other programs (logger, WSJT-X…).
- **Serial port** and speed; advanced parameters if needed (bits, stop, flow control, forced DTR / RTS). The list shows every active Windows port (Prolific, CH340, Silicon Labs and virtual ports included); you can also type the port name, e.g. COM9.
- **PTT method**: CAT (recommended), DTR or RTS (also on the port of an interface next to the radio), or VOX.
- **Transmit audio source**: “Rear / Data” (USB audio, ACC) or “Front / Mic”. Choose “Front / Mic” if your radio transmits USB audio without DATA mode: HFImage then leaves its mode alone.

- **Mode**: “Data / Pkt” puts the radio in DATA before each transmission, keeping its sideband (LSB → LSB-DATA, USB → USB-DATA).
- **Test CAT** reads the frequency; **Test PTT** transmits for 1.5 s. Green = OK, red = error shown below.

Radio not in DATA before transmitting

As soon as CAT is connected, HFImage reads the radio's mode before each transmission, **whatever the PTT method** (CAT, RTS, DTR or VOX). If it is not in DATA, the setting “**Radio not in DATA before transmitting**” (Settings, below the radio buttons) decides:

- **Switch to DATA automatically (USB-DATA / LSB-DATA)**: the default. The radio goes to DATA **keeping its sideband**: in LSB (40 and 80 m, as in EasyPal) it goes to **LSB-DATA**, in USB to **USB-DATA**. The log shows it (“TS-890S switched to LSB-DATA1 by CAT”).
- **Ask**: a window offers to switch to DATA, transmit anyway or cancel.
- **Transmit without asking**: the radio stays in its mode; the log reminds you at each transmission.

Kenwood TS-990: a single command (LSB-D1 or USB-D1). TS-890, TS-590, TS-480: LSB or USB, then DATA. Other radios: PKTLSB / PKTUSB (Hamlib), DIG-L / DIG-U (OmniRig), FLRig's DATA mode. With RTS, DTR or VOX PTT, a failed switch to DATA is only reported in the log and does not block the transmission.

Kenwood: Kenwood TS-890 / TS-990: DATA audio source = USB Audio, **DATA VOX OFF**, CAT PTT on the radio's virtual COM port at 115200 baud, source “Rear / Data”. The radio switches to LSB-DATA or USB-DATA by itself. The port must be free: close ARCP or your logger, or go through OmniRig. If the input and output of the USB codec have the same name in Windows, HFImage always picks the right one.

Layout: separate windows

Menu **Windows**: tick an area to turn it into a **separate window**, placed and sized wherever you want, even on another screen: **panel and buttons**, **waterfall**, **stations heard**, **Stations** tab and **gallery**. The RX, TX and Simulation tabs and the log stay in the main window.

- Separate windows always stay **in front of** the main window and are minimised with it.
- Untick the area, or close its window, to put it back. “Put everything back in the main window” gathers everything; “Settings → Original window layout” too.
- The layout is kept from one session to the next.
- **Show the history** and **Show received messages** (RX tab) hide these lists: without both, the received picture uses the whole width.

4. Receiving pictures

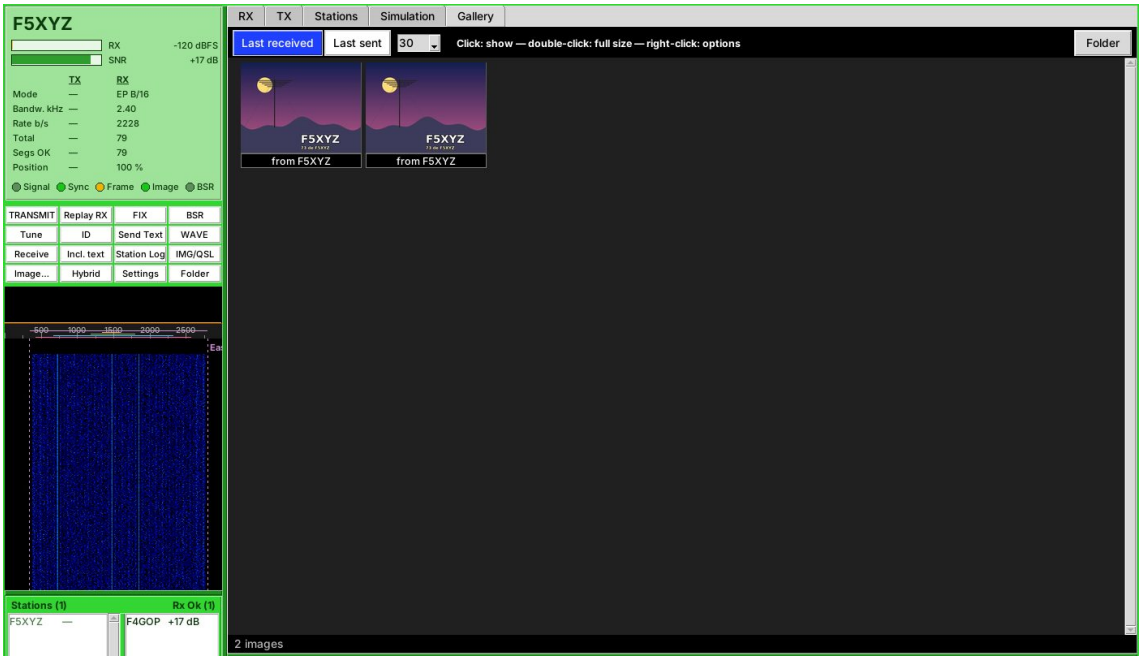
Listening starts by itself when the audio input is set (**Receive** button to stop / resume). Nothing to choose: **all modes are decoded at the same time** — DATAC1X, DATAC1, DATAC3, DATAC4 **and** EasyPal (HamDRM), whatever mode is selected for transmission.

- **Status panel** (top left): callsign of the station received, radio frequency and mode, RX level, SNR, mode, bandwidth, data rate, segments received, Signal / Sync / Frame / Image / BSR lights.
- When a picture starts to arrive while the **TX tab** is displayed, HFImage switches to the **RX tab** so that you see it come in (except during your own transmissions).

- **Quick preview** (HFImage modes): a small picture arrives in ~20 s, then the full picture replaces it.
- **History** on the right: click to show again, double-click for full size (can be hidden, menu Windows).
- **Zoom**: mouse wheel on the picture to enlarge / reduce around the pointer (up to ×16), hold the left button to move the enlarged picture, right-click to return to the whole picture.
- **Automatic BSR / ACK** (HFImage modes): if pieces are missing at the end, HFImage asks for them (BSR); the sender sends only what is missing (FIX). Once the picture is complete, an acknowledgement (ACK) with your report goes back to the sender.
- Pictures are stored in the chosen folder (Settings → Received images folder); **Folder** button to open it.
- **Replay RX**: decodes the last 15 minutes again (if a picture was missed).

Gallery

Gallery tab: the last 30, 60 or 100 pictures received (“from F4EMK”) or sent. Click: show; double-click: full size; right-click: send the picture again, reply with this picture, show the station on the map, open the folder, remove from the gallery.



The gallery of the last pictures received.

5. Sending a picture (HFImage modes)

1. **TX** tab → **Open image…** (JPG, PNG, GIF, WebP, AVIF, SVG, BMP…).
2. Choose the **Mode** (Auto recommended), **Max duration** (120 s by default), **Width** (auto), **Format** (avif recommended) and **Robustness** (Auto recommended).
3. HFImage computes the size and quality that fit in the duration: the compressed picture is shown with its size and quality.
4. ▶ **Transmit** (or TRANSMIT button). The progress bar follows the transmission.
5. If stations ask for missing pieces (BSR), **FIX** sends only what is missing (automatically by default).

Auto mode and robustness

In **Auto** mode, HFImage chooses the mode from the **weakest report** of the stations that received your pictures during the last 45 minutes (Rx Ok list). Without reports it uses the SNR of the stations heard, otherwise DATAC1X. The reason for the choice is shown under the picture.

SNR shown (report)	Mode chosen	Pictures
9 dB and more	DATAC1X	sharpest
5 to 9 dB	DATAC1	medium
below 5 dB	DATAC3	small but very robust

Robustness is the share of spare frames (fountain code): they replace the frames lost in QSB. In “Auto”: 15 % if the link has margin, 25 % if it is tight, 40 % if it is at the limit. More robustness = slightly less detail for the same duration, but the picture gets through in one go.

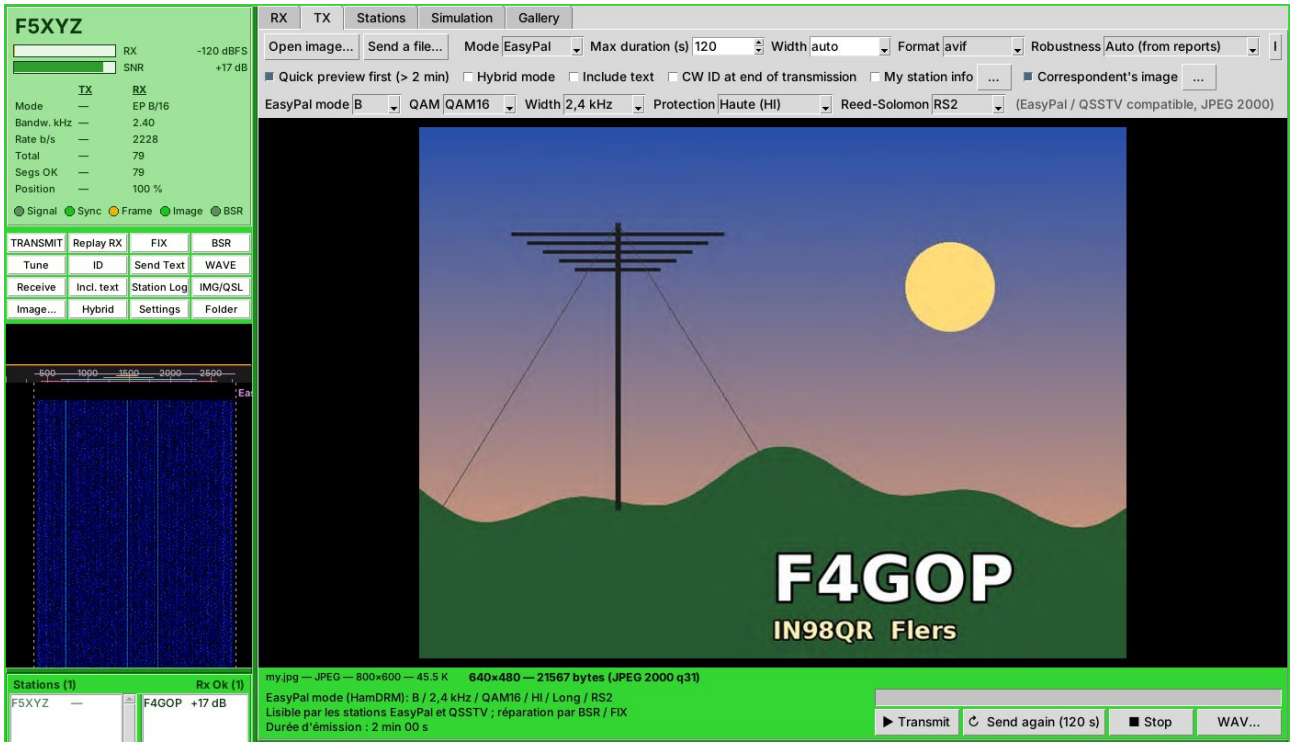
Tip: reports and SNR are the **real SNR over 3 kHz**, comparable from one mode to another. On a very strong signal the measurement saturates: reports are capped at +15 dB in DATAC3 / DATAC0 and +21 dB in DATAC1X / DATAC1. For a sharper picture, the most effective is a longer duration (3 min in DATAC1X ≈ 50 % more bytes).

The HFImage modes

Mode	Width	Net rate	In 2 min	Use
DATAC1X	2.2 kHz	~1,500 bit/s	~13-16 kB	Good conditions: sharp pictures (HFImage only)
DATAC1	1.7 kHz	~980 bit/s	~8-10 kB	Average conditions
DATAC3	0.6 kHz	~320 bit/s	~2.5-3 kB	Weak signals, QSB
DATAC4	0.25 kHz	~85 bit/s	~1 kB	Very weak signals
DATAC0	0.6 kHz	—	—	BSR, ACK, presence (automatic)

6. EasyPal mode (HamDRM) — EasyPal / QSSTV compatible

Choose **Mode: EasyPal** in the TX tab. A second row appears with the EasyPal settings, the same as in EasyPal. The signal is the HamDRM signal of EasyPal and QSSTV (DRM OFDM): those stations decode your pictures, and HFImage decodes theirs.



TX tab in EasyPal mode: EasyPal settings row, compressed picture (JPEG 2000) and exact transmission time.

EasyPal settings

Setting	Values	Meaning
EasyPal mode	A, B, E	Robustness of the OFDM signal. A: fastest, good conditions. B: usual HF mode. E: most robust (long paths, strong QSB).
QAM	QAM4, QAM16, QAM64	Modulation: QAM4 robust and slow, QAM64 fast but needs a strong signal.
Width	2.2 kHz, 2.4 kHz	Bandwidth, labelled as in EasyPal (2.3 / 2.5 kHz in QSSTV, same signal). Use 2.2 kHz with a narrow filter.
Protection	High (HI), Normal (LO)	Code rate of the error correction. HI is more robust.
Reed-Solomon	None, RS1 … RS4	Extra redundancy: the picture is rebuilt even when segments are lost. **RS2 by default**. Read by EasyPal and QSSTV.

Pictures are sent in **JPEG 2000** (the usual EasyPal format) or JPEG, sized to fit the chosen duration; the exact transmission time is computed before transmitting. The waterfall marks the exact band of the mode received (A / B / E, 2.2 / 2.4 kHz).

Recommended settings

Conditions	Settings	SNR needed (about)
Good propagation, strong signal	A / QAM64	15 dB
Usual HF (most common)	B / QAM16 / RS2	10 dB without QSB, 14 dB with light QSB
Difficult propagation, QSB	B or E / QAM4 / RS3 or RS4	6 dB without QSB, about 12 dB with average QSB

Measured in the simulator: with light QSB, B / QAM16 with **RS2** gives a complete picture from about 14 dB instead of 17-20 dB without Reed-Solomon, for a transmission about 18 % longer. HFImage also transmits the EasyPal signal with a reduced crest factor (+4 dB average level for the same peak power) and decodes it with an improved bit-reliability estimate (about +0.5 dB in QAM16) — the signal remains 100 % EasyPal compatible.

Receiving EasyPal pictures

- The status panel shows the callsign, “EP B/16” (EasyPal mode / QAM), bandwidth, data rate, total and received segments, position, and the Sync / Frame / Image / BSR lights.
- If a picture remains incomplete, the BSR light turns amber and the log tells you to click **BSR**: the EasyPal BSR asks the sender for the missing segments.
- Texts sent by EasyPal / QSSTV (.txt files) appear in the received messages window with the callsign.

Answering BSR (FIX)

When an EasyPal or QSSTV station asks for missing segments of your last picture (BSR), HFImage answers **automatically** with a FIX containing only those segments (option in the Stations tab), or manually with the **FIX** button.

Texts in EasyPal mode

Send Text in EasyPal mode sends the message as a .txt file, readable by EasyPal and QSSTV.

Hybrid EasyPal

With **Hybrid mode** ticked in EasyPal mode, HFImage works like EasyPal: the picture is uploaded by FTP to **<folder>/HybridFiles1/de_CALL-n-name.jpg**, and only a short announcement is sent over the air, in mode **E / QAM4** (set automatically, and restored when you untick hybrid mode). EasyPal, QSSTV and HFImage stations that receive it download the picture automatically and send their Rx Ok with report.

Important: as in EasyPal, the announcement contains the FTP server, user name and password **scrambled, not encrypted**: any station receiving it can read them. Use a **dedicated FTP account** limited to one folder. Only plain ASCII characters are allowed in these fields.

Following the received mode

With “**Switch transmission to the received mode**” ticked (Stations tab, on by default), the transmission mode changes as soon as a picture or message starts to arrive in the other family of modes:

- EasyPal received → mode **EasyPal**; when the picture is complete, the A / B / E mode, width and QAM of the station received are also taken over (except for hybrid announcements,

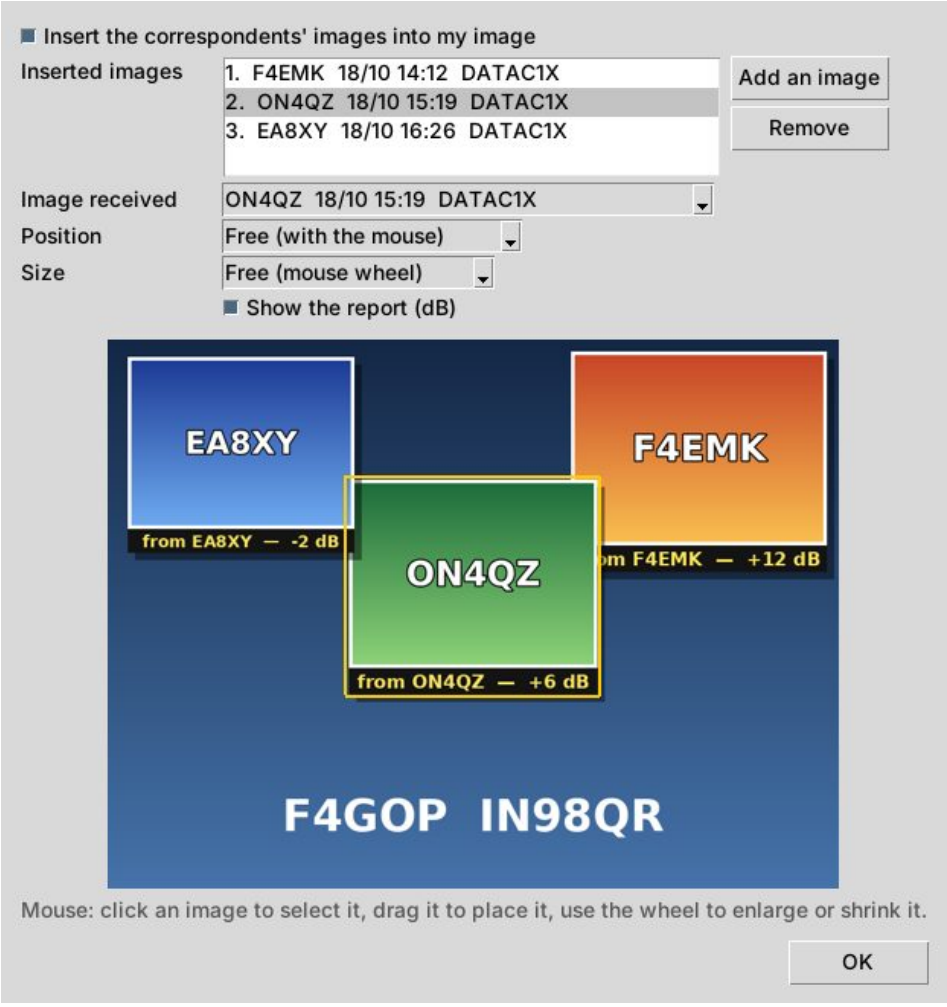
always E / QAM4);
- HFImage (DATAC) received while in EasyPal → mode **Auto**.
- Nothing changes during your own transmissions or when decoding a WAV file. Each change is written in the log.

7. Adding to your picture

Correspondent's picture

To answer as in EasyPal: right-click a picture in the **Gallery** → “Reply with this image (inserted into my TX image)”, or tick **Correspondent's image** in the TX tab and click **…**. Received pictures are inserted into yours with the caption “from F4EMK — +12 dB”: the **report** is the SNR you measured when receiving them. **Up to 4 pictures** at once, for example to answer several stations together.

- **Inserted images** list: **Add an image** (it goes to a free corner) and **Remove**. The “Image received” list changes the picture of the selected line.
- Each picture has its own **position** and **size**: a corner and a size in the lists, or with the mouse in the preview. **Click** a picture to select it (yellow frame), **drag** it to place it, the **mouse wheel** enlarges or reduces it (10 % to 90 % of the width).
- Gallery, right-click: “**Reply with this image**” replaces the insets with this picture only; “**Add this image to my TX image**” adds it to the others.
- The layout is kept; the picture transmitted is exactly the preview.



Correspondent's image: three pictures inserted, ON4QZ's selected (yellow frame).

My station info

Tick **My station info** (TX tab) and click **…** next to it (or Images menu): a banner is inserted at the bottom or top of the transmitted picture, with your callsign, locator, QTH, radio (read by CAT if the field is empty), power, antenna, frequency, UTC date and time, and free text. Each item can be ticked or not; the preview updates live. The banner is part of the picture, so every correspondent sees it, whatever program they use. The **Font and colours** frame sets the font, bold / italic, text colour, callsign colour, outline, background and its opacity, and a coloured border; **Presets…** offers ready-made styles.

Other transmissions

- **Incl. text**: a message attached to the picture, shown in “Received messages”.
- **Send Text**: message alone.
- **Send a file…** (HFImage modes): any file (text, PDF, ZIP, ADIF log…), compressed when useful and protected by the same fountain code (BSR / FIX and ACK included); with hybrid mode, the file goes via the Internet and only the link goes over the air. Received files are stored under their original name in the **fichiers** subfolder.
- **Tune** (10 s): carrier to adjust the amplifier / tuner. **ID**: callsign in Morse.
- **WAVE** / waterfall text: begin.wav / end.wav files, **text or picture** drawn in the correspondent's waterfall (Action → Waterfall text…); “Save” makes it a begin.wav played before each transmission, as in EasyPal.
- **IMG/QSL**: QSL card with background photo, callsign, locator and report filled in automatically, font and colours of your choice.

8. Messages

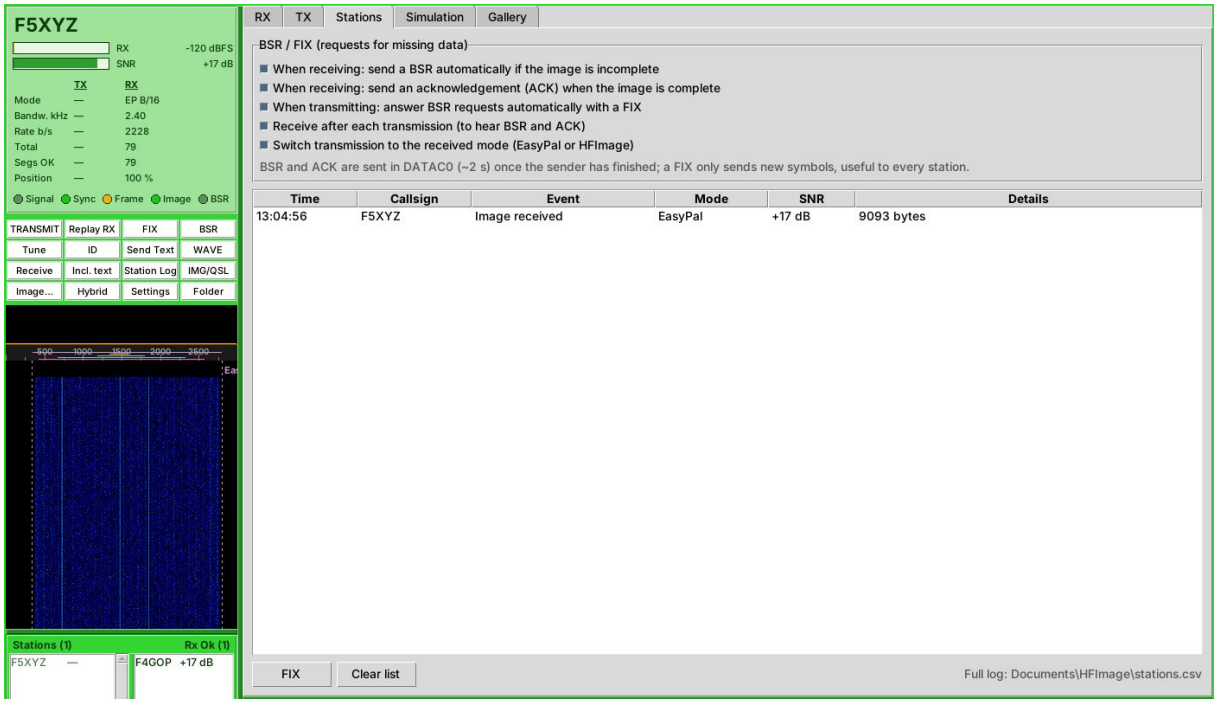
When a correspondent sends a message (text alone, or text attached to a picture), the **Message received** window opens in front with a beep: callsign, UTC time, mode, SNR, the text in large print and the thumbnail of the picture if there is one (click = full size). The ◀ ▶ arrows browse the messages of the session. **Reply…** opens the text window pre-filled with “F1BAL de F4GOP: ”. The sender's callsign is also written above each message in the “Received messages” frame. That frame can be hidden (menu Windows): messages still arrive in the “Message received” window.

Did my correspondent receive my message? As soon as a message is received, the correspondent's program sends an acknowledgement, over the air (ACK in DATAC0) and via the Internet (if the network is on). In the **Send Text** window, the list “Sent messages — received by” shows “✓ F1BAL (+8 dB)” in green when it arrived, “waiting for acknowledgement…” in orange, then “no acknowledgement” in red after 3 minutes.

9. Stations online, Rx Ok and map

With **Settings → Internet** ticked, each running HFImage announces itself on a public server (callsign, locator, frequency, transmitting / receiving): the **Stations** list immediately shows the stations that have HFImage open, with their frequency. No other data is shared.

- In the list, * marks a station transmitting (red) and **HYB** a station in hybrid mode (blue). Double-click a station for details in the log.
- **Rx Ok**: the stations that received your picture, with their **report in dB**, visible to everyone (over the air and via the Internet). Also works for EasyPal pictures and hybrid EasyPal pictures received by HFImage stations.
- Click a station: the **station map** (like Digi-Sites), with path and distance; “Paste the map into the TX window” to send it as a picture.
- Firewall or Internet provider blocking port 1883: HFImage tries the other accesses of the same server by itself (8883 encrypted, then 8884 and 8000 via the web); the log shows which one worked.



Stations tab: BSR / FIX options, mode following, and the station log.

Hybrid mode (HFImage) and website

In the HFImage modes, hybrid mode sends the picture through your **web hosting** (FTP); only the **link** goes over the air (~11 s in DATAC3). Stations download it automatically. Ideal for large pictures and **animated GIFs**. Settings → Hybrid mode: FTP server, user name, password, folder and **public link** of this folder; “Test server” checks everything. Only the link goes over the air, never the password.

Flash hybrid: tick **Flash (5 s)** next to “Hybrid mode” (TX tab). The announcement no longer carries the link, only the picture's **fingerprint** (shortened SHA-256), size and format: one **DATAC0** frame (500 Hz) sent 5 times, **4.5 s** instead of 11 s. The link goes at the same time through the online-stations server; the receiving station downloads the picture and checks it has **exactly the fingerprint received over the air**. Simulated: fading −2 dB, 85 to 90 % of announcements received against 55 to 60 % for the normal announcement. Both stations need “**Stations online**” (Settings → Internet) and version 0.71 or later; if the server is unreachable, HFImage sends the normal announcement.

Download safety: an announcement received over the air can only fetch a file of **8 MB at most**, and only from a public Internet address (never the computer itself or a machine on your home network). Received pictures and files always stay in the reception folder, whatever the announced callsign.

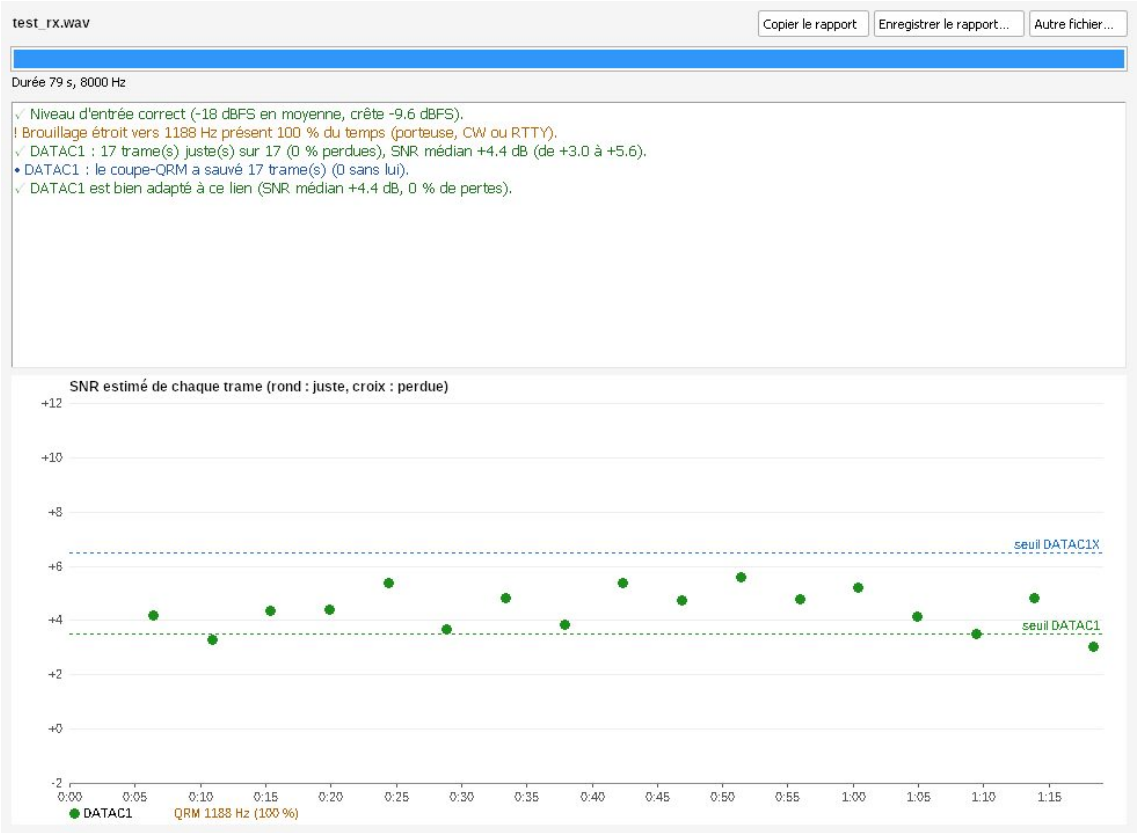
Menu **Images → Website / Digi-Sites…**: as in EasyPal, each received picture is uploaded as **image1.jpg** (the previous ones move to image2, image3…), with callsign, date, mode and band; your transmitted pictures go to TX1.jpg … TX5.jpg, and a **hfimage.html** page shows them. Example with qsl.net: server **ftp.qsl.net**, empty folder, link **https://www.qsl.net/yourcall**, page for Digi-Sites **https://www.qsl.net/yourcall/hfimage.html**.

10. Noise, QSB and QRM

- **Noise and QSB** (HFImage modes): Auto mode + Auto robustness. If losses come in groups, choose the “Strong (70 %)” robustness.
- **Noise and QSB** (EasyPal mode): keep Reed-Solomon (RS2 or more), lower the QAM (QAM4) or use mode E. An incomplete picture can be completed with BSR / FIX.
- **QRM**: the **QRM cutter** (Settings → Audio) removes, before demodulation, everything outside the band of the mode (adjacent SSB) and, inside the band, narrow lines (carrier, CW, RTTY).
- A **wide interferer on top of the signal** (another SSB station on the same frequency) cannot be removed: move the frequency.

Reception analyser

To understand a failed reception: **WAVE** button → ● **Record RX** during the reception, then **Files → Analyse a WAV recording…**. HFImage reports the input level (clipping, too low), narrow interference and its frequency, the frames received / lost in each mode, SNR, frequency offset between the radios, the type of losses (QSB or isolated), what the QRM cutter saved, and the recommended mode. A recording can also be decoded again with **Files → Decode a WAV file** (HFImage and EasyPal).



The analyser: plain-language diagnosis and SNR of each frame (screenshot of the French interface).

11. Common problems

Symptom	Probable cause	What to do
The radio does not transmit	PTT badly set, COM port busy	Radio… → Test PTT; close the programs holding the port or use OmniRig
The radio transmits without power	Radio not in DATA, wrong audio source, level too low	“Switch to DATA automatically”; source “Rear / Data”; raise the level (ALC must not move)
“Radio not in DATA” and it does not switch	Setting “Transmit without asking” or “Ask”	Settings → “Radio not in DATA before transmitting” → “Switch to DATA automatically”
COM port missing from the list	Driver not seen, or device unplugged	Version 0.76 or later; type the port name (COM9); check Windows Device Manager
Nothing is received	Wrong audio input, Windows volume muted	Settings → Audio; check that the waterfall moves; analyse a WAVE recording
Frames detected but lost	Weak signal, QSB, QRM, clipping	WAV analyser; Auto mode / Strong robustness; lower the input level
EasyPal picture incomplete	QSB, segments lost	Click BSR; ask the sender to use RS2 or more, QAM4 or mode E
HFImage picture incomplete	BSR not heard or no FIX	Check “Send BSR” on your side and “answer BSR with a FIX” on the sender's side (Stations tab)
Pictures not sharp	Slow mode or strong robustness, short duration	Auto robustness; 3 min duration; DATAC1X or EasyPal B/QAM16 if reports allow
No stations online	Internet unticked or server blocked	Settings → Internet; HFImage tries ports 8883, 8884, 8000 by itself; see the log
Flash hybrid: picture not downloaded	“Stations online” off at the sender or receiver	Enable Internet on both sides, or untick “Flash (5 s)”
Hybrid mode: FTP failure	Credentials, folder or public link	Settings → Hybrid mode → Test server

Full diagnosis: **HFImage_F4GOP.exe --selftest report.txt** (modems including EasyPal, map, Internet, sound cards, COM ports, radio). Log: **Documents\HFImage\journal.txt**.

Credits and licence

The EasyPal (HamDRM) modem comes from **QSSTV** (Johan Maes ON4QZ), itself derived from **Dream** (Volker Fischer, TU Darmstadt) and **RXAMADRM** (Ties Bos PA0MBO), GNU GPL v3. The DATAC modes use **codec2** (David Rowe VK5DGR and contributors). Tribute to **Erik Sundstrup VK4AES**, author of EasyPal.

Contact

F4GOP — Denis — f4gop.denis@gmail.com
Program page: github.com/f4gopdenis-cyber/HFImage_F4GOP (versions, changes, sources).
73!